

CY6P01 Research Project

Quantification and Characterisation of Microplastic Release from Commercial Chewing Gums

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Abstract

Chewing gum is one of the most widely consumed products globally. Its synthetic polymer gum base, which contains materials such as polyisobutylene, polyvinyl acetate, and ethylene-vinyl acetate, has received comparatively little attention as a potential source of microplastic oral exposure.

In this study, five commercial chewing gums were analysed following incubation in artificial saliva with potassium hydroxide digestion, followed by filtration, light microscopy and ATR-FTIR spectroscopy. Between 6 and 9 suspected microplastic particles were identified per sample (3.50-6.47 particles g⁻¹), with no particles detected in the blank control. Polymer identity was confirmed by ATR-FTIR spectroscopy: polypropylene (PP) in Samples 1 and 4, ethylene-vinyl acetate (EVA) in Sample 3, and polyvinyl chloride (PVC) in Sample 5, with a tentative PVC assignment in Sample 2. Spectral assignments were supported by characteristic absorption features (including the diagnostic ester carbonyl at 1733 cm⁻¹ for EVA and C–Cl stretching bands at 716/671 cm⁻¹ for PVC) as well as OpenSpecy database match values of 0.72-0.78.

Particle yields were substantially lower than values reported by studies using mechanical chewing simulation, consistent with the static digestion protocol underestimating true release. All five samples nonetheless released detectable polymeric material confirmed by spectroscopy, providing evidence that chewing gum is a source of oral microplastic exposure warranting further investigation.

Further work using mechanical simulation and analysis of multiple particles per sample is needed to better estimate real-world exposure and polymer composition.

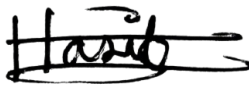
Key words: microplastics, chewing gum, ATR-FTIR, polypropylene, PVC, ethylene-vinyl acetate, oral exposure, polymer characterisation

Acknowledgements

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Plagiarism Statement

I, Mohammad Hasibul Haque, hereby confirm that the work submitted in this project is my own. Any ideas submitted in this project are my own. Any ideas, quotations, and paraphrasing from other people's work and publications have been appropriately referenced. I have not violated London Metropolitan University's policy on plagiarism.

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Abbreviations

ATR	Attenuated Total Reflectance
EVA	Ethylene-Vinyl Acetate copolymer
FTIR	Fourier Transform Infrared Spectroscopy
KOH	Potassium Hydroxide
MP	Microplastic
PE	Polyethylene
PP	Polypropylene
PVC	Polyvinyl Chloride
PVA	Polyvinyl Acetate
SNR	Signal-to-Noise Ratio
WHO	World Health Organization

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1. Introduction

1.1 Microplastics: Definition and Prevalence

Microplastics are defined as plastic particles smaller than 5 mm in their longest dimension. They have emerged as a significant environmental and public health concern over the past two decades (Thompson et al., 2004; EFSA, 2016). Once considered primarily a marine pollution problem, microplastics are now detected in freshwater systems, agricultural soils, indoor and outdoor air, and a wide range of human food and consumer products (Barboza et al., 2018; Zhang et al., 2021; Dris et al., 2017). More recently, microplastics have been identified in human blood, lung tissue, placenta, and faeces, providing evidence that exposure and internal accumulation occur across multiple biological compartments (Leslie et al., 2022; Jenner et al., 2022). The scale of this exposure has prompted the WHO (2022) to identify microplastic ingestion through food and drinking water as a growing public health concern and to call for standardised methodologies and further exposure assessment.

Microplastics are broadly categorised as primary and secondary. Primary microplastics are intentionally manufactured at small scale, such as microbeads in cosmetics and personal care products. Secondary microplastics come from the fragmentation of larger plastic items through mechanical wear, UV degradation, or chemical breakdown (Cole et al., 2011; Rochman et al., 2021). A third, often ignored category includes particles generated from synthetic consumer products during normal use, such as particles shed from synthetic textiles during washing (de Falco et al., 2019) and particles abraded from vehicle tyres (Panko et al., 2013). This third category also includes particles released from the synthetic polymer gum base of commercial chewing gums during chewing, as examined in this study.



Figure 1. Classification of microplastics by source. Primary microplastics originate from intentionally manufactured small-scale plastic items (microbeads, plastic pellets, cosmetic additives, and textile fibres), while secondary microplastics arise from the unintentional degradation of larger macroplastic items. Adapted from Barboza et al. (2018).

1.2 Chewing Gum Composition and Microplastic Potential

Global chewing gum consumption is substantial. The market was valued at approximately USD 32.6 billion in 2021 and is projected to grow steadily, with billions of pieces consumed annually worldwide (Fortune Business Insights, 2022). Unlike traditional gums derived from natural chicle, the vast majority of commercial gums manufactured since the mid-twentieth century rely on wholly synthetic polymer gum bases. These bases typically comprise synthetic elastomers including polyisobutylene, polyisoprene, and polybutadiene along with resin polymers such as polyvinyl acetate (PVA) and terpene resins. These bases also contain plasticisers including glycerol and lanolin derivatives (Lebling, 2018; Morales-Caselles et al., 2021). Ethylene-vinyl acetate (EVA) copolymers have also been reported as common gum base constituents and fall within the definition of microplastic materials under current regulatory frameworks.

The sustained mechanical action of chewing forms a potential mechanism for the breakdown and release of synthetic polymer particles from the gum base matrix into saliva. Ruiz-Aceituno et al. (2021) first quantified this release systematically, reporting between 37 and 637 microplastic particles per gram of gum using a mechanical chewing simulator, identifying PP, PE, and EVA as predominant polymer types. Lowe and Koelmans (2019) confirmed comparable particle counts across multiple brands under simulated chewing conditions; however, both studies used mechanical simulators calibrated to specific chewing frequencies and forces, which introduces variability depending on the protocol used and may not fully represent individual chewing behaviour. Neither study directly compared mainstream branded products with cheaper retail alternatives, nor included an off-brand product. Despite this growing evidence base, chewing gum is currently excluded from microplastic exposure regulations and food labelling requirements in the UK and EU.

1.3 FTIR Spectroscopy for Microplastic Identification

ATR-FTIR spectroscopy is widely regarded as a gold-standard technique for the chemical characterisation of suspected microplastic particles (Araujo et al., 2018; Cowger et al., 2021). The technique operates on the principle that molecular bonds absorb infrared radiation at specific wavenumbers corresponding to their vibrational frequencies. In the attenuated total reflectance (ATR) configuration, infrared radiation is directed through an internal reflection element (IRE) which is typically a diamond, germanium, or zinc selenide crystal at an angle that produces total internal reflection. At the point of total reflection, an evanescent wave extends a short distance (0.5–3 μm) beyond the crystal surface into the sample pressed against it. Molecular bonds within this penetration depth absorb radiation at their characteristic wavenumbers, and the resulting spectrum of attenuated reflectance is measured. The method works by measuring the differential absorption of infrared radiation by molecular bonds, including C–H, C–C, C=O, and C–Cl bonds, at characteristic wavenumbers. This produces a spectral fingerprint unique to each polymer class. The ATR configuration requires minimal sample preparation, allows analysis of intact solid particles, and can be applied to particles as small as 10–20 μm (Käppler et al., 2016). In this study, spectra were acquired using an Agilent Technologies Cary 630 ATR-FTIR spectrometer, a compact benchtop instrument (14 × 26 × 16 cm; 3.5 kg) with permanently aligned optics and a diamond ATR crystal, capable of a spectral resolution of $\leq 2\text{ cm}^{-1}$ and a spectral range of 350–7,000 cm^{-1} (KBr). The instrument was operated using MicroLab Expert control software. Spectral libraries such as the OpenSpecy open-access database (Cowger et al., 2021) allow quantitative matching against reference spectra, with match values above 0.70 generally considered indicative of a positive polymer identification.

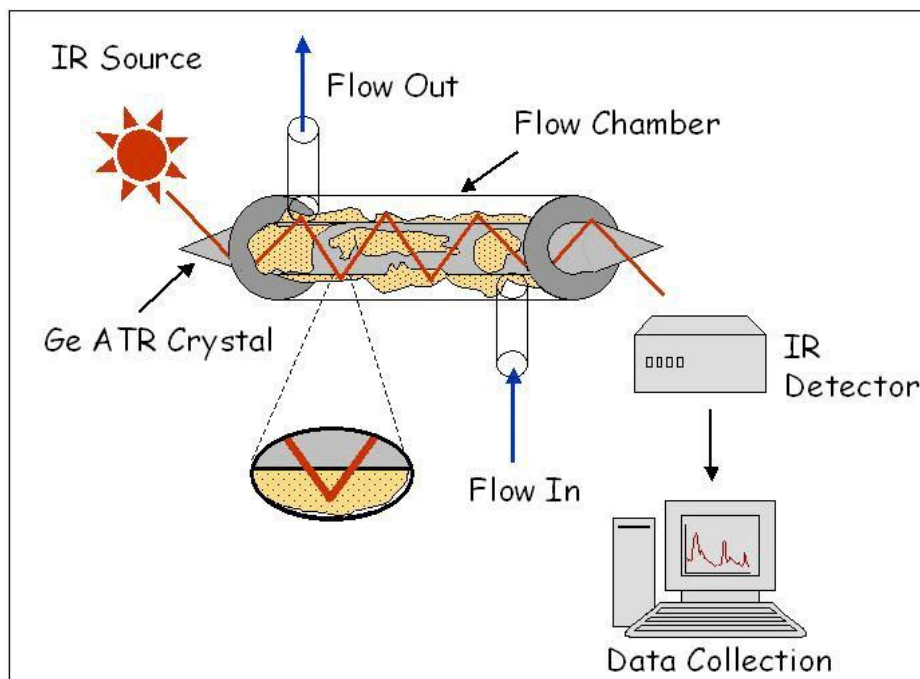


Figure 2. Schematic diagram of an ATR-FTIR flow-cell configuration showing a germanium (Ge) ATR crystal. Infrared (IR) radiation from the source enters the crystal and undergoes multiple total internal reflections. At each reflection point, an evanescent wave penetrates a short distance (0.5–3 μm) into the sample in the flow chamber. Molecular bonds within this depth absorb characteristic wavenumbers; the attenuated beam exits to the IR detector and the resulting spectrum is recorded by the data collection system. Adapted from K  ppler et al. (2016).

1.4 Rationale and Study Objectives

Researchers increasingly recognise chewing gum as a source of microplastics, yet comprehensive studies comparing different brands using multiple analytical methods remain rare. No published study has examined differences in microplastic release between mainstream branded gums and cheaper retail alternatives. This study addresses that gap by examining five commercially purchased gum products: four Wrigley’s branded varieties and one off-brand product from Primark. A KOH-saliva digestion protocol with vacuum filtration was used, followed by light microscopy and ATR-FTIR spectroscopy to quantify and chemically characterise any particles released.

2. Aims

The primary aim of this project was to determine whether synthetic polymer microplastics are released from commercial chewing gums under conditions simulating the oral environment, and to characterise the polymer types present.

The specific aims were:

1. To apply a chemical digestion and filtration protocol for the extraction of particles from commercial chewing gums, using artificial saliva and KOH solution, and to document procedural challenges and adaptations made during the study.
2. To quantify the number of suspected microplastic particles released per sample using light microscopy, and to normalise particle counts to gum mass (particles per gram).
3. To confirm the chemical identity of recovered particles using ATR-FTIR spectroscopy with OpenSpecy database matching.
4. To compare microplastic release across five gum products, including branded and off-brand alternatives, and discuss findings in the context of human oral microplastic exposure.

The working hypothesis was that all five gum products would release particles consistent with synthetic polymers, and that FTIR analysis would identify polymer types in line with known gum base composition.

3. Results and Discussion

Five commercial chewing gum products were each incubated in a mixture of artificial saliva and 10% KOH solution at 37°C for one hour, followed by 24 hours of further digestion at room temperature. This treatment was designed to dissolve organic and biological matrix components while leaving synthetic polymer particles intact. Following digestion, each sample was separated into a liquid supernatant fraction and a solid pellet fraction, and both were independently filtered through 11 µm pore-size filter papers under vacuum. The resulting dried filter papers were examined by light microscopy for particle enumeration, and selected particles were subsequently transferred to an ATR-FTIR instrument for chemical identification. A blank control (no gum) was processed in parallel throughout.

3.1 Microscopic Particle Counts and Quantification

Microscopic examination of all ten filter papers (supernatant and pellet fractions for each sample) revealed particles consistent with synthetic microplastics in every case. No particles were observed in either control filter paper, indicating that detected particles originated from the gum material rather than reagent contamination or airborne deposition. Particle counts and normalised values are presented in Table 1.

S#	Brand	Mass (g)	Supernatant	Pellet	Total	Per gram	Polymer (FTIR)
1	Wrigley's Juicy Fruit	1.25	5	1	6	4.80	PP
2	Wrigley's Apple Extra	1.46	5	4	9	6.16	PVC (tentative)
3	Wrigley's Airwaves Extreme	1.39	7	2	9	6.47	EVA
4	Wrigley's Extra Refreshers	2.00	1	6	7	3.50	PP
5	Off-brand Watermelon (Primark)	1.43	2	5	7	4.90	PVC
C	Control (no gum)	-	0	0	0	-	-

Table 1. Microplastic particle counts from supernatant and pellet fractions, total particle count, normalised particle yield (particles g⁻¹ of gum), and polymer identity confirmed by ATR-FTIR. S# = sample number; PP = polypropylene; PVC = polyvinyl chloride; EVA = ethylene-vinyl acetate; C = control.

Total particle counts ranged from 6 (Sample 1, Wrigley's Juicy Fruit) to 9 (Samples 2 and 3). When normalised to gum mass, particle yields ranged from 3.50 particles g⁻¹ (Sample 4, Wrigley's Extra Refreshers, 2.00 g) to 6.47 particles g⁻¹ (Sample 3, Wrigley's Airwaves Extreme, 1.39 g). The highest absolute count per gram was recorded for Sample 3, the lightest piece in the study, suggesting that particle yield per unit mass is relatively consistent across the Wrigley's product range but may be influenced by gum base density and piece thickness. Sample 5 (off-brand Primark, 1.43 g) yielded 4.90 particles g⁻¹, within the range observed for branded products, indicating that particle release is not substantially higher in low-cost alternatives under these extraction conditions. However, the qualitative polymer class (PVC versus PP for Wrigley's products) does differ between branded and off-branded gums.

The distribution between supernatant and pellet fractions varied markedly between samples. Samples 1, 2, and 3 returned most particles from the supernatant (5, 5, and 7 particles respectively), suggesting

that lower-density or more hydrophilic polymer particles were dispersed into the liquid phase during incubation. In contrast, Samples 4 and 5 yielded more particles from the pellet (6 and 5 respectively), indicating that denser or more adherent polymer fragments remained associated with the solid gum residue. This pattern is broadly consistent with the polymer densities involved: PP has a density of approximately 0.90 g cm^{-3} and would be expected to float in the aqueous digestion medium, while PVC has a density of approximately $1.30\text{-}1.45 \text{ g cm}^{-3}$ and would be expected to sediment. Samples 4 and 5 were identified as PP and PVC respectively, yet Sample 4 (PP) produced more pellet particles, which may reflect particle entrapment within the solid gum residue rather than true density-driven sedimentation. These distribution differences may also reflect variation in particle size or gum base matrix cohesion, and no formal mechanistic conclusion can be drawn from these counts alone.

3.2 Microscopic Observations

Representative microscope images captured during particle enumeration are shown as Figures 4-9. Particles exhibited two principal morphologies: filamentous fibres and irregular fragments consistent with the microplastic categories described in the literature (Lusher et al., 2017; Zhang et al., 2021). Blue and black fibre-type particles were distinctly visible against the white filter paper background (Figures 4, 5, 8, 9). Several samples contained both prominent coloured fibres and finer translucent filaments (Figure 7). Figure 6 shows a large yellowish-green film-like deposit with an embedded dark fibre, consistent with a composite polymer structure and with the EVA identification subsequently obtained for Sample 3.

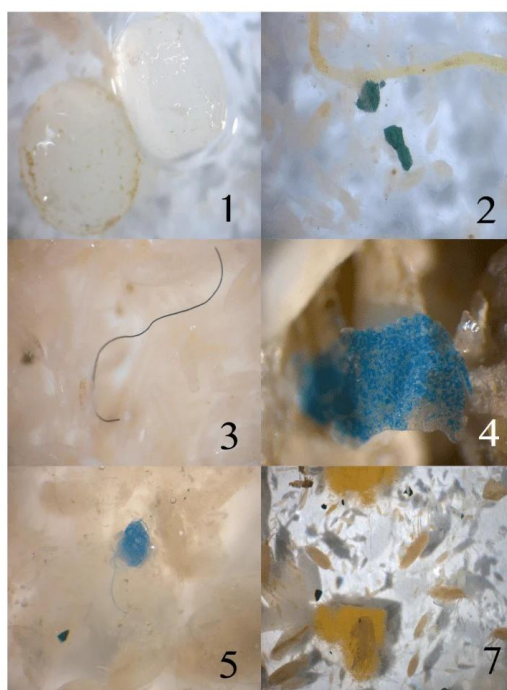


Figure 3. Representative light microscopy images of microplastic morphologies recovered from chewing gum samples in the present study. Images 1 and 2 show pellet and fragment morphologies; Images 3-5 show fibre and film morphologies including a blue synthetic fragment; Image 7 shows a mixed deposit containing fragments and films of varying colour. Particle categories follow the classification scheme of Lusher et al. (2017).



Figure 4. Light microscope image showing a prominent blue synthetic fibre alongside multiple fine translucent filaments and irregular white fragments, typical of particles recovered from gum supernatant fractions.

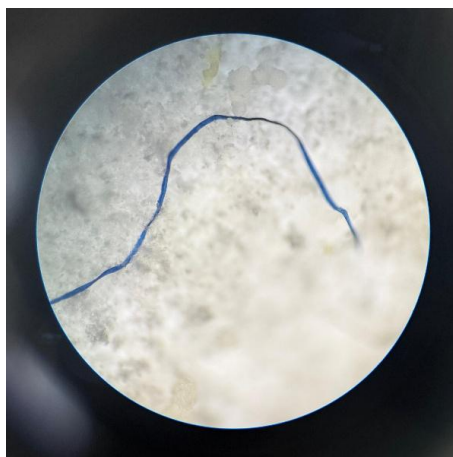


Figure 5. Light microscope image showing a single clearly defined blue polymer fibre with uniform morphology across the filter paper background, consistent with a synthetic microfilament.

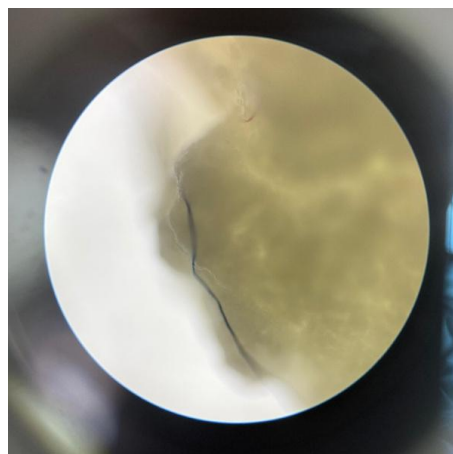


Figure 6. Light microscope image showing a large yellowish-green film-like polymer deposit with an embedded dark fibre running along its edge. The two-phase appearance is consistent with a composite EVA polymer structure.



Figure 7. Light microscope image showing dark filamentous fibres distributed within a network of finer translucent filaments, suggesting the presence of multiple polymer fibre types on a single filter paper section.

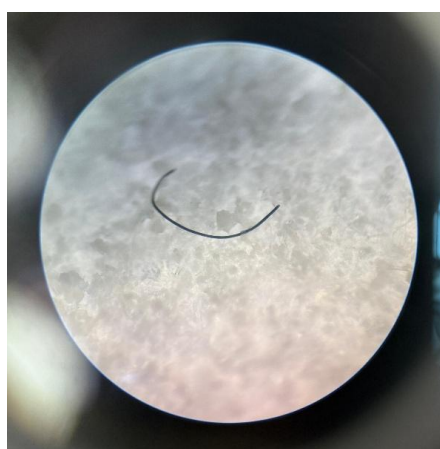


Figure 8. Light microscope image showing a curved black fibre with C-shaped morphology, consistent with a fragment of synthetic polymer thread separated during mechanical or chemical processing.



Figure 9. Light microscope image showing a short hooked dark fibre alongside a small granular aggregate, consistent with an abraded polymer microfibre and associated particulate residue.

3.3 ATR-FTIR Spectral Analysis

ATR-FTIR spectra were acquired for particles from all five samples using the Agilent Technologies Cary 630 ATR-FTIR instrument (32 scans, resolution 8 cm⁻¹, range 4000-650 cm⁻¹, Happ-Genzel apodization). For each sample, the full instrument output - including the spectrum and complete peak table is presented below, followed by the OpenSpecy database comparison. A consolidated summary of FTIR results is given in Table 2.

Sample	Brand	Material Class	Spectrum Match	Match	SNR
1	Wrigley's Juicy Fruit	Polypropylene (PP)	PP 25. White cutting board	0.77	22.57
2	Wrigley's Apple Extra	Polyhaloolefins (PVC)	PVC.1	0.74	7.14
3	Wrigley's Airwaves Extreme	Ethylene-vinyl acetate (EVA)	EVA 2. Yellow fragment	0.78	9.22
4	Wrigley's Extra Refreshers	Polypropylene (PP)	Polypropylene	0.72	24.36
5	Off-brand Watermelon (Primark)	Polyhaloolefins (PVC)	PVC.1	0.72	18.78

Table 2. Summary of ATR-FTIR spectral analysis results for all five gum samples. Match value and signal-to-noise ratio (SNR) derived from OpenSpecy database comparison (openanalysis.org/openspecy/). All spectra acquired on 21 April 2026.

Sample 1 - Wrigley's Juicy Fruit (Polypropylene)

The full ATR-FTIR instrument output for Sample 1 is shown in Figure 10. Eleven peaks were identified, with principal absorptions at 2918.5 cm^{-1} and 2851.4 cm^{-1} (C–H asymmetric and symmetric stretching of methyl/methylene groups), 1420.1 cm^{-1} (C–H bending), 1107.0 cm^{-1} and 1017.6 cm^{-1} (C–C stretching/skeletal vibrations), and a fingerprint peak at 711.9 cm^{-1} . A carbonyl-region absorption at 1736.9 cm^{-1} is also noted, which may reflect a minor oxidation product or co-contaminant. These spectral features are characteristic of an aliphatic hydrocarbon polymer consistent with polypropylene.

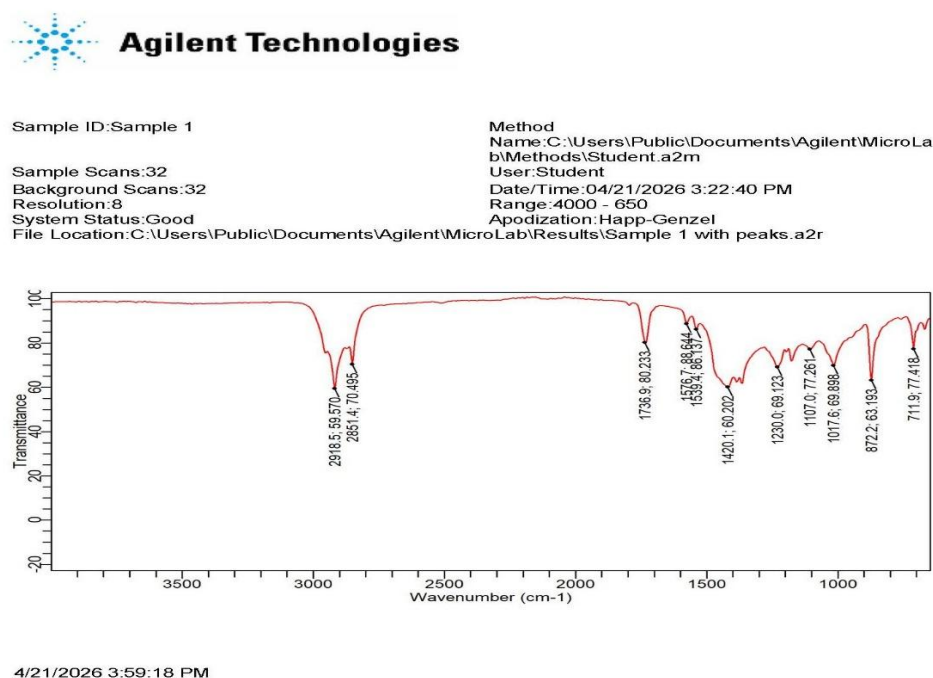


Figure 10. Full ATR-FTIR instrument output for Sample 1 (Wrigley's Juicy Fruit), Agilent Technologies Cary 630 ATR-FTIR, 21/04/2026 15:22. Transmittance (%) vs. wavenumber (cm^{-1}); 32 scans; resolution 8 cm^{-1} ; range $4000\text{--}650\text{ cm}^{-1}$. Complete peak table shown below spectrum.

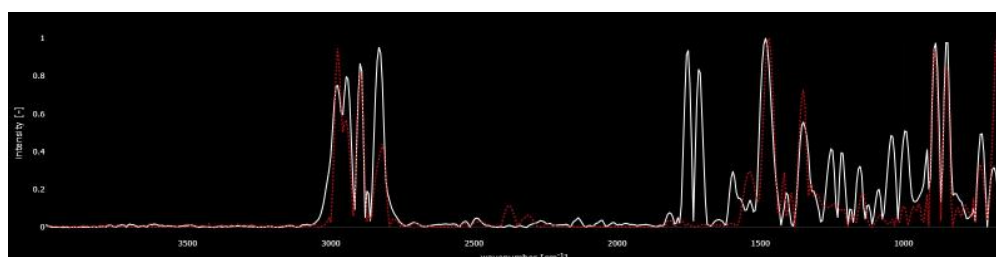


Figure 11. OpenSpecy database spectral comparison for Sample 1. Solid white line = obtained ATR-FTIR spectrum; red dashed line = reference (PP 25. White cutting board fragment). Match value: 0.77; SNR: 22.57. Material class: Poly(polypropylene).

Sample 2 - Wrigley's Apple Extra (Polyvinyl Chloride - tentative)

The full ATR-FTIR output for Sample 2 is shown in Figure 12. Transmittance values exceed 90% at nearly all recorded peaks, indicating shallow sample absorption depth - consistent with the low SNR of 7.14. Peaks were identified at 2918.5 and 2851.4 cm^{-1} (C–H stretching), 1736.9 cm^{-1} , 1576.7 and 1543.1 cm^{-1} (C–C/C=C ring or chain absorptions), and multiple peaks in the 870-1420 cm^{-1} fingerprint region. While the OpenSpecy comparison yielded a match value of 0.74 against PVC.1 (Figure 13) - above the commonly applied 0.70 threshold - the poor spectral quality necessitates that this identification be considered tentative pending confirmation with a more strongly absorbing particle from this sample.

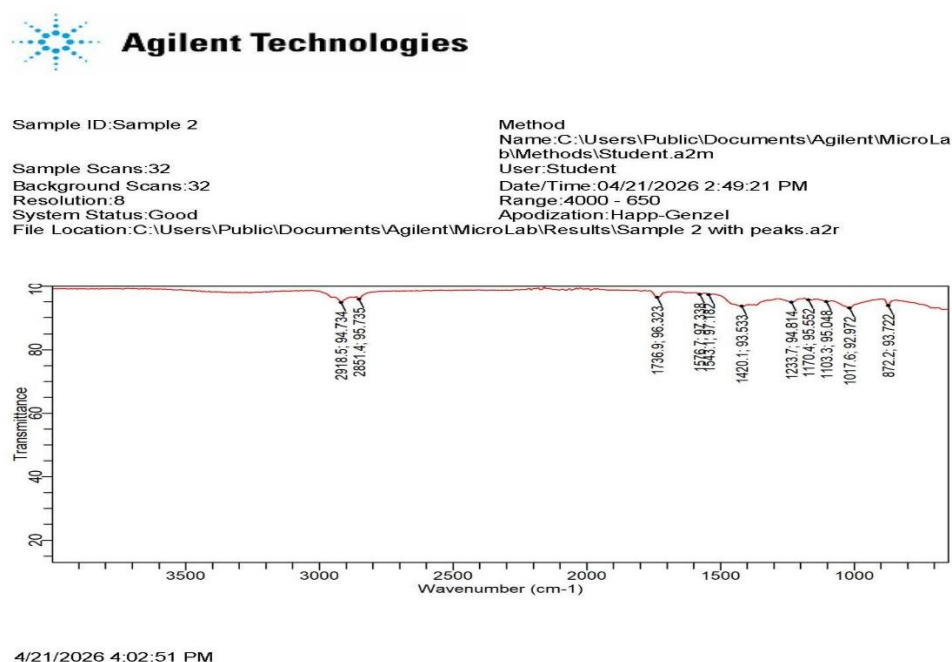


Figure 12. Full ATR-FTIR instrument output for Sample 2 (Wrigley's Apple Extra), Agilent Technologies Cary 630 ATR-FTIR, 21/04/2026 14:49. Note characteristically high transmittance values (>90% at most peaks), indicating shallow absorption depth and low spectral signal quality.

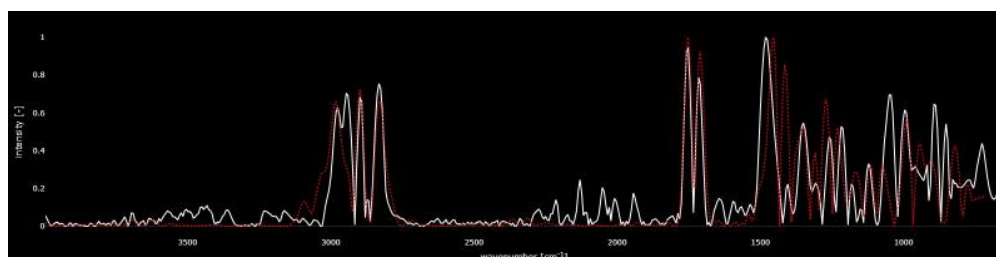


Figure 13. OpenSpecy database spectral comparison for Sample 2. Solid white line = obtained ATR-FTIR spectrum; red dashed line = reference (PVC.1). Match value: 0.74; SNR: 7.14. Material class: Polyhaloolefins (vinylhalides). Identification tentative due to low SNR.

Sample 3 - Wrigley's Airwaves Extreme (Ethylene-Vinyl Acetate)

The full ATR-FTIR output for Sample 3 is shown in Figure 14. Eleven peaks were recorded, with notable absorptions at 2922.2 and 2855.1 cm^{-1} (C–H stretching), a strong ester carbonyl absorption at 1733.2 cm^{-1} - characteristic and highly diagnostic for the acetate ester groups of EVA copolymers - 1423.8 cm^{-1} , 1367.9 cm^{-1} , 1177.8 cm^{-1} (C–O–C ester stretching), and 667.2 cm^{-1} . The combination of a prominent carbonyl band at $\sim 1733 \text{ cm}^{-1}$ with C–O stretching in the 1100–1250 cm^{-1} region is diagnostic for vinyl acetate ester functionality and constitutes strong evidence for EVA identification independent of the database match.

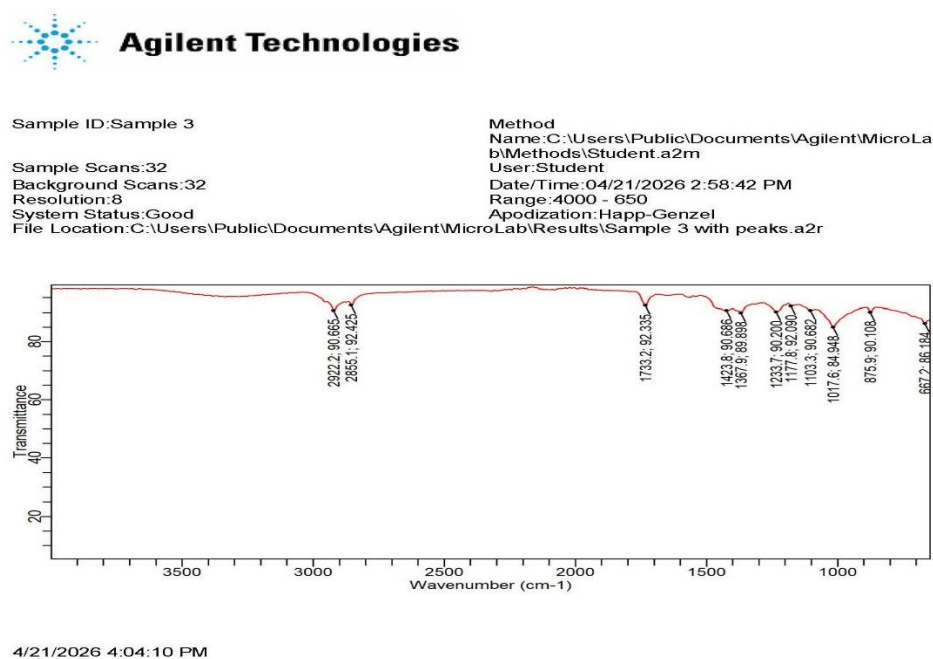


Figure 14. Full ATR-FTIR instrument output for Sample 3 (Wrigley's Airwaves Extreme), Agilent Technologies Cary 630 ATR-FTIR, 21/04/2026 14:58. Note the characteristic ester carbonyl absorption at $\sim 1733 \text{ cm}^{-1}$ and C–O stretching bands in the 1100–1250 cm^{-1} region, diagnostic for EVA copolymers.

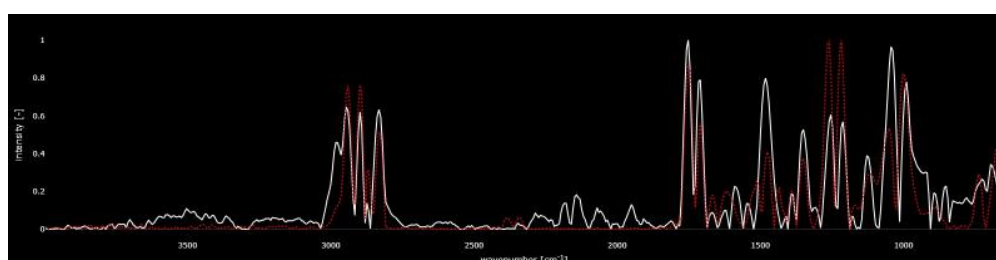


Figure 15. OpenSpecy database spectral comparison for Sample 3. Solid white line = obtained ATR-FTIR spectrum; red dashed line = reference (EVA 2. Yellow fragment). Match value: 0.78 (highest in study); SNR: 9.22. Material class: Ethylene-vinyl acetate.

Sample 4 - Wrigley's Extra Refreshers Peppermint (Polypropylene)

The full ATR-FTIR output for Sample 4 is shown in Figure 16. Twelve peaks were identified, including a diagnostic C–H stretching triplet at 2955.8, 2918.5, and 2851.4 cm^{-1} - a pattern characteristic of polypropylene, arising from asymmetric and symmetric stretching of $-\text{CH}_3$ and $-\text{CH}_2$ groups. Additional absorptions at 1461.1 and 1386.6 cm^{-1} (C–H bending/deformation), 1233.7 cm^{-1} , and 715.6 cm^{-1} further support this assignment. The SNR of 24.36 is the highest in the study, indicating excellent spectral quality and providing the most reliable polymer identification among all samples.

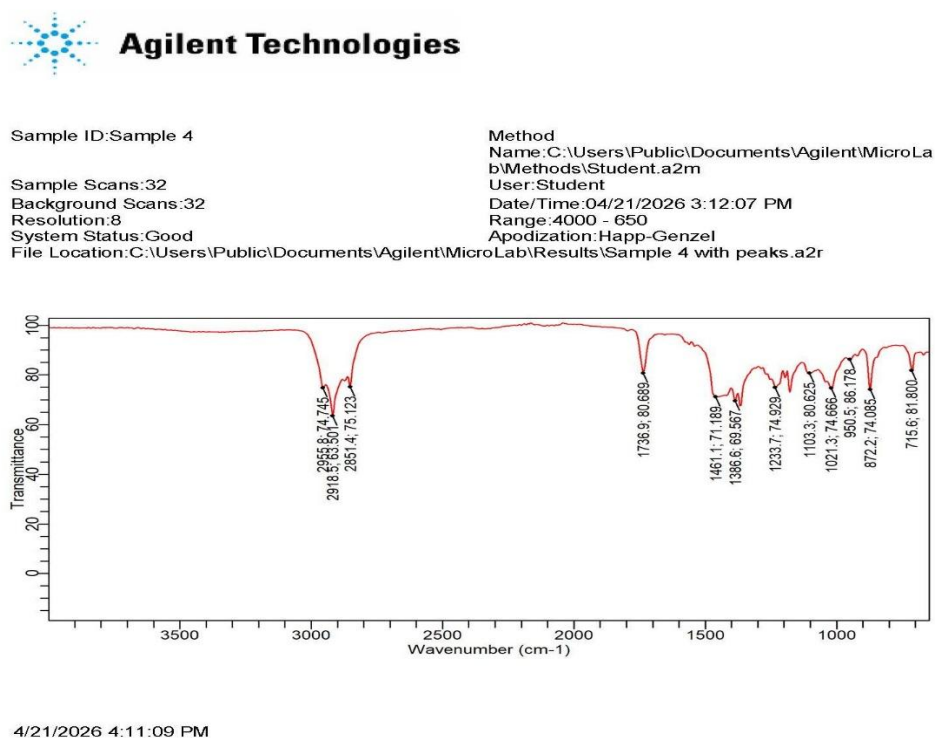


Figure 16. Full ATR-FTIR instrument output for Sample 4 (Wrigley's Extra Refreshers Peppermint Sugarfree), Agilent Technologies Cary 630 ATR-FTIR, 21/04/2026 15:12. The C–H stretching triplet at $\sim 2956/2918/2851 \text{ cm}^{-1}$ and bending bands at $\sim 1461/1386 \text{ cm}^{-1}$ are characteristic of polypropylene. SNR 24.36 - highest spectral quality in study.

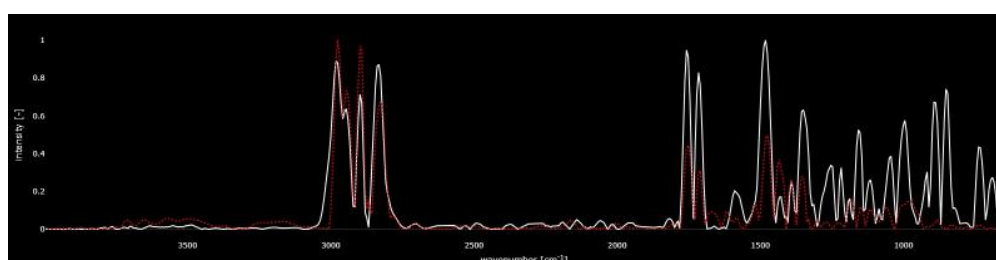


Figure 17. OpenSpecy database spectral comparison for Sample 4. Solid white line = obtained ATR-FTIR spectrum; red dashed line = reference (Polypropylene). Match value: 0.72; SNR: 24.36. Material class: Poly(polypropylene).

Sample 5 - Off-brand Watermelon Gum, Primark (Polyvinyl Chloride)

The full ATR-FTIR output for Sample 5 is shown in Figure 18. Twelve peaks were recorded, including absorptions at 2955.8, 2918.5, and 2851.4 cm^{-1} (C–H stretching), 1736.9 cm^{-1} , 1420.1 and 1386.6 cm^{-1} , 1233.7 cm^{-1} , 1110.7 cm^{-1} , and notably dual low-frequency peaks at 715.6 and 670.9 cm^{-1} . This pair of absorptions in the 650–720 cm^{-1} region is consistent with C–Cl stretching vibrations characteristic of polyvinyl chloride, providing spectroscopic evidence for PVC that complements the OpenSpecy match. The SNR of 18.78 is substantially higher than that of Sample 2 (also matched to PVC), and the PVC assignment for Sample 5 is accordingly more reliable.

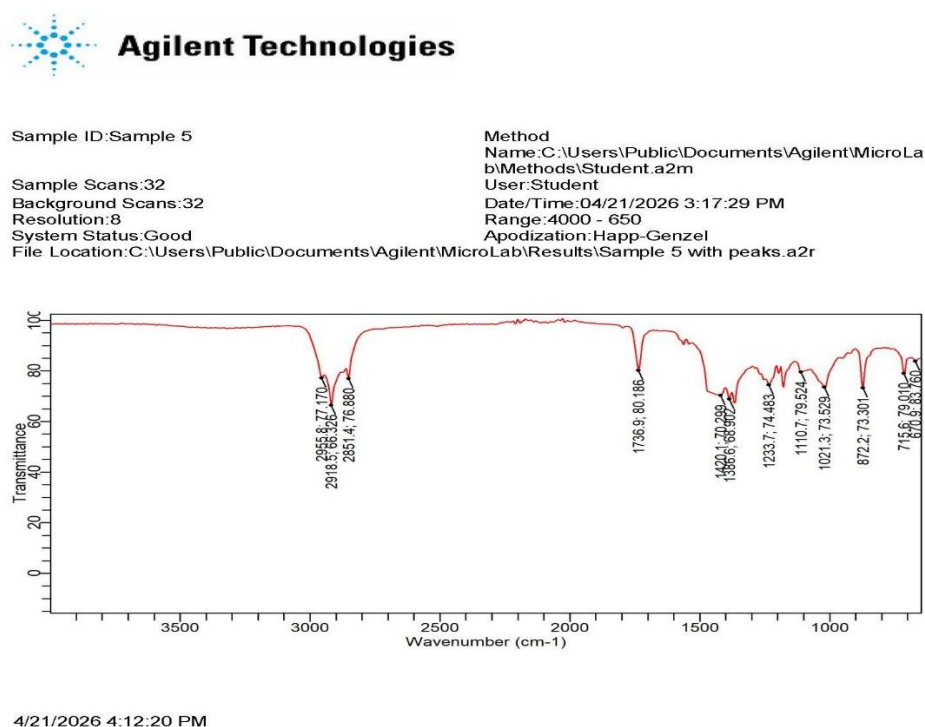


Figure 18. Full ATR-FTIR instrument output for Sample 5 (Off-brand Watermelon Gum, Primark), Agilent Technologies Cary 630 ATR-FTIR, 21/04/2026 15:17. Dual low-frequency peaks at ~ 716 and ~ 671 cm^{-1} are consistent with C–Cl stretching in polyvinyl chloride.

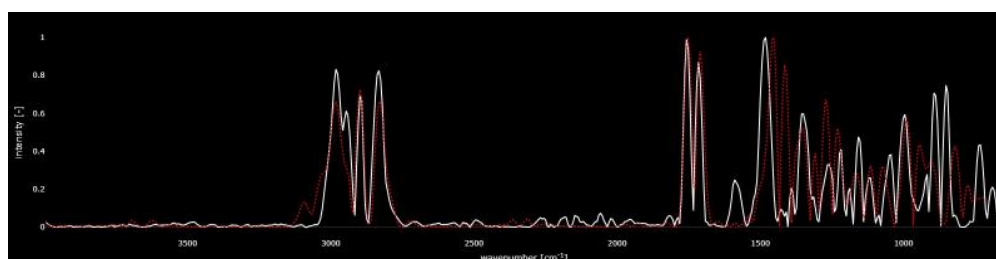


Figure 19. OpenSpecy database spectral comparison for Sample 5. Solid white line = obtained ATR-FTIR spectrum; red dashed line = reference (PVC.1). Match value: 0.72; SNR: 18.78. Material class: Polyhaloolefins (vinylhalides).

3.4 Comparison with Published Literature

The particle counts recorded in this study (3.50–6.47 particles g⁻¹) are substantially lower than those reported in mechanistic studies using chewing simulators. Ruiz-Aceituno et al. (2021) documented 37–637 particles g⁻¹ across nine brands, and Lowe and Koelmans (2019) reported comparable counts across additional brands, both employing mechanical chewing action. It is worth noting that the wide range reported by Ruiz-Aceituno et al. (2021), nearly 20-fold between lowest and highest values, suggests that polymer composition, gum base density, and piece mass all influence release rate. A single particle count per brand, as in the present study, is therefore unlikely to capture this variability. Regardless, the difference between the present results and those of both mechanical studies is expected and reflects a fundamental methodological distinction. The static KOH-saliva chemical digestion used here does not replicate the shear forces generated during active chewing, which are the primary physical driver of polymer particle release from the gum base matrix. The present particle counts therefore represent a conservative lower bound on true *in vivo* particle generation. No statistical comparison was performed due to the limited sample size (n = 5 products, n = 1 FTIR analysis per sample).

The polymer types identified in this study (PP, PVC, and EVA) are consistent with those reported in the wider literature. Ruiz-Aceituno et al. (2021) identified PP and EVA as major components of gum-derived microplastics from nine commercial brands, and similar aliphatic polyolefins have been reported by Lowe and Koelmans (2019). The detection of EVA in Sample 3 is directly consistent with its known role as a synthetic elastomer and texture modifier in commercial gum bases (Lebling, 2018; Morales-Caselles et al., 2021). PP identified in Samples 1 and 4 likely reflects its use as a processing resin or plasticiser carrier in the gum base matrix, as PP-type particles have previously been attributed to resin binders in chewing gum formulations (Ruiz-Aceituno et al., 2021). PVC is not listed as a primary ingredient in published gum base formulations; its presence in Samples 2 and 5 most plausibly reflects its use as a polymeric plasticiser carrier resin or a processing equipment-derived contaminant, consistent with previous reports of PVC-type materials in gum residues (Morales-Caselles et al., 2021). The identification of PP across two different Wrigley's product lines, and PVC in both a Wrigley's product and the Primark off-brand product, is consistent with the widespread industry practice of using standardised synthetic gum base formulations across multiple flavour variants.

3.5 Toxicological Implications

The findings of this study have potential public health implications, though they must be interpreted within the constraints of the dataset. As a purely illustrative order-of-magnitude exercise, and acknowledging that the static digestion method substantially underestimates true particle release, the measured yield of 3.50–6.47 particles g⁻¹ would, assuming complete particle ingestion, correspond to roughly 2,500–4,700 particles per year for a consumer chewing two pieces daily (Fortune Business Insights, 2022). This should be understood as a conservative lower bound, not an exposure estimate, given the methodological limitations described in Section 3.6. For context, Cox et al. (2019) estimated total annual microplastic ingestion from all dietary sources at 74,000–121,000 particles for adults, and chewing gum has not been systematically included in such models to date. The primary value of these figures is to illustrate that, even under conservative conditions, gum-derived particles are not negligible relative to other documented sources.

The identification of PVC is of toxicological concern. Commercial PVC formulations routinely incorporate phthalate plasticisers. This include di(2-ethylhexyl) phthalate (DEHP) and dibutyl phthalate which are classified as endocrine-disrupting substances and are restricted under EU Regulation 10/2011 in food contact materials. Whether the PVC-type particles identified here contain such additives, and whether these migrate into saliva during the sustained contact of chewing, remains unknown and constitutes a priority research question. The WHO (2022) has highlighted the need to assess not only the physical presence of microplastic particles but also the chemical hazard posed by associated additives as a component of risk assessment.

The health effects of ingested microplastics at realistic human exposure levels remain an area of active investigation. Leslie et al. (2022) demonstrated the presence of microplastics in human blood at concentrations suggesting ongoing systemic exposure, and Jenner et al. (2022) identified microplastics in human lung tissue. Animal and in vitro studies have identified potential inflammatory, oxidative stress, and barrier disruption effects (Barboza et al., 2018; Rochman et al., 2019), though direct extrapolation to human health outcomes at current exposure levels remains uncertain. Regulatory frameworks have yet to address chewing gum specifically, representing a gap that the present findings support addressing.

3.6 Methodological Limitations

Several limitations of the present study must be acknowledged. First, the use of static chemical digestion without mechanical chewing simulation means that particle yields substantially underestimate real-world release. Future studies should employ a validated mechanical chewing simulator to generate physiologically representative particle counts. Second, light microscopy without fluorescent staining (e.g., Nile Red) limits detection to particles visible at the magnifications employed, and particles smaller than approximately 50 μm may have passed through the filter paper pores or been too small to identify visually. The addition of fluorescence microscopy would substantially improve detection efficiency and enable size distribution characterisation.

Third, FTIR analysis was performed on only one particle per sample, which limits conclusions about within-sample polymer heterogeneity. A minimum of five particles per sample is recommended for future work to enable statistical characterisation of polymer diversity. The absence of a positive polymer control (e.g., a certified PP or PVC standard processed through the same protocol) also prevents formal validation of the particle recovery and identification pipeline. The tentative PVC identification for Sample 2, driven by poor spectral quality (SNR 7.14), illustrates the importance of particle selection and sample transfer technique in ATR-FTIR analysis. Finally, an earlier FTIR session in March 2026 was unsuccessful due to particle loss during transfer, requiring protocol modification; the final method employed partially protruding particles to minimise this risk, but a proportion of analysable particles may still have been lost across both sessions.

4. Conclusions and Future Work

This study has demonstrated that synthetic polymer microparticles were detected in all five commercial chewing gum products tested under simulated oral conditions, with at least one polymer particle confirmed per sample by ATR-FTIR spectroscopy. Between 6 and 9 particles were recovered per sample (3.50-6.47 particles g^{-1}), with zero particles from the blank control, confirming gum as the source. ATR-FTIR spectroscopy confirmed polypropylene (PP) in Samples 1 and 4, ethylene-vinyl acetate (EVA) in Sample 3, and polyvinyl chloride (PVC) in Sample 5, with OpenSpecy match values of 0.72-0.78; Sample 2 returned a tentative PVC assignment (match value 0.74) that should be treated with caution given the low spectral signal-to-noise ratio (SNR 7.14). The raw instrument spectra provide independent spectroscopic evidence supporting these assignments: the diagnostic ester carbonyl at $\sim 1733 \text{ cm}^{-1}$ for EVA (Sample 3), the C–H stretching triplet at $\sim 2956/2918/2851 \text{ cm}^{-1}$ for PP (Samples 1 and 4), and the dual C–Cl stretching absorptions at $\sim 716/671 \text{ cm}^{-1}$ for PVC (Sample 5). These polymer identities are consistent with the known synthetic composition of commercial gum bases: EVA is a documented gum base elastomer and texture modifier (Lebling, 2018); PP is consistent with its use as a resin binder or processing additive previously identified in gum-derived particles by Ruiz-Aceituno et al. (2021); and PVC, while not a primary listed ingredient, has been attributed to polymeric plasticiser carrier resins in prior analyses of gum residues (Morales-Caselles et al., 2021). The findings therefore reflect expected gum base polymer chemistry rather than incidental contamination.

These findings contribute to the growing evidence base identifying chewing gum as a source of oral microplastic exposure that remains outside the scope of current UK and EU food labelling and microplastic exposure regulation. The identification of PVC in Sample 5 is potentially the most toxicologically significant finding of this study. PVC formulations routinely contain phthalate plasticisers, which are classified as endocrine-disrupting substances and restricted under EU Regulation 10/2011. This finding requires confirmation, as only one particle was analysed per sample and no additive migration testing was carried out. Particle yields measured here are conservative lower bounds; true release during active mastication is expected to be substantially higher.

Future work should include mechanical chewing simulation to generate more physiologically relevant particle yields. Improved detection methods such as fluorescence microscopy with Nile Red staining could help identify smaller particles below $50 \mu\text{m}$. Polymer characterisation would also benefit from analysing multiple particles per sample using FTIR, alongside techniques such as SEM-EDX for combined morphological and elemental analysis. In addition, chemical migration testing is needed to assess potential release of plasticiser additives from PVC-associated particles. Finally, expanding the study to a wider range of brands, flavours, and gum types would improve the generalisability of the findings. Together, these advances would strengthen the evidence base needed to inform regulation on polymer composition disclosure in commercial chewing gum.

5. Experimental

5.1 General Experimental

All laboratory work was conducted at London Metropolitan University. Artificial saliva and KOH solutions were prepared freshly prior to each experimental run. Glassware was cleaned with distilled water and allowed to dry before use to minimise contamination. Vacuum filtration was performed using a standard Buchner funnel assembly. Light microscopy was performed using a Cole-Parmer MSU-200 compound binocular brightfield microscope. ATR-FTIR spectra were recorded on an Agilent Technologies Cary 630 ATR-FTIR spectrometer equipped with a diamond ATR crystal, using MicroLab software (Student method profile). Spectral database matching was performed using the OpenSpecy web platform (openanalysis.org/openspecy/, Cowger et al., 2021).

5.2 Reagent Preparation

Artificial saliva (250 mL) was prepared by dissolving NaCl (2.0 g), KCl (0.05 g), NaH_2PO_4 (0.195 g), and Na_2HPO_4 (0.09 g) in approximately 200 mL of distilled water. The initial pH of 5.6 was adjusted to 6.95 by dropwise addition of additional Na_2HPO_4 solution. The solution was made to 250 mL with distilled water and stored in a labelled glass bottle. This formulation approximates the principal inorganic ionic composition of human saliva (Dawes, 2008). A 10% w/v KOH solution (500 mL) was prepared by dissolving 50 g of KOH pellets in approximately 200 mL of distilled water with stirring, allowing the exothermic dissolution to cool to room temperature, then making to 500 mL. KOH was used to digest proteinaceous and biological matrix material, selectively retaining synthetic polymer particles.

5.3 Gum Sample Preparation and Incubation

Five commercial chewing gum products were purchased from UK retail outlets and stored in original packaging at room temperature until use. Individual pieces were weighed on an analytical balance before processing: Sample 1- Wrigley's Juicy Fruit (1.25 g); Sample 2- Wrigley's Apple Flavour Extra (1.46 g); Sample 3- Wrigley's Airwaves Extreme (1.39 g); Sample 4- Wrigley's Extra Refreshers Peppermint Sugarfree (2.00 g); Sample 5- Off-brand watermelon flavour gum from Primark (1.43 g). Each piece was placed in a 100 mL glass beaker and 15 mL each of artificial saliva and KOH solution were added (30 mL total). A blank control beaker received 15 mL each of saliva and KOH without gum. All beakers were incubated in a water bath at 37°C for one hour, then sealed with clingfilm and aluminium foil and rested at room temperature for 24 hours to complete KOH digestion of organic constituents.

5.4 Filtration and Sample Processing

Following the 24-hour digestion period, the liquid supernatant was carefully decanted from each beaker into a separate labelled vessel. The remaining solid pellet (undissolved gum residue) was gently disrupted with a clean glass rod where necessary to release embedded particles. Both supernatant and pellet fractions were filtered independently through Whatman QL100 filter papers (0601A00006; pore size 11 μm) under vacuum using a Buchner funnel assembly. Each filter paper was carefully transferred using clean forceps to a labelled petri dish, covered, and left to air-dry at room temperature. The paired supernatant and pellet filter papers for each sample were stored separately until microscopic examination. Two earlier filtration attempts (27 February and 25 March 2026) were unsuccessful due to filter paper rupture under vacuum pressure; the final protocol used reduced vacuum pressure and careful paper placement to prevent recurrence.

5.5 Microscopic Examination

Each dried filter paper was cut in half using clean scissors. Each half was fixed to a glass microscope slide using small adhesive tape strips at the corners to prevent movement. Slides were examined using a Cole-Parmer MSU-200 compound binocular brightfield microscope equipped with a set of semi-plan objectives at 4×, 10×, 40×, and 100× magnification, wide-field eyepieces (WF10×/18 mm), and a 130 × 130 mm mechanical stage with 70 × 28 mm X-Y translation. The instrument is powered via an external 100-240 V mains adapter. The filter paper surface was scanned systematically under transmitted brightfield illumination. Particles were recorded when they were visually distinguishable from the filter paper background by colour, morphology, or three-dimensional relief. Particles consistent with synthetic microplastic morphology characterised by uniform colouration (blue, black, or translucent), sharp linear or curved geometry, and non-biological appearance were counted and photographed. Control slides were examined first. All particle counts were made by the same observer in a single session to minimise inter-observer variability.

5.6 ATR-FTIR Spectroscopic Analysis

ATR-FTIR analysis was performed using the Agilent Technologies Cary 630 instrument (32 sample scans; 32 background scans; resolution 8 cm⁻¹; range 4000-650 cm⁻¹; Happ-Genzel apodization). The diamond ATR crystal was cleaned with isopropanol and a blank background spectrum collected immediately before each sample measurement. Particles selected for analysis were chosen based on partial protrusion from the filter paper surface to allow scraping with a fine implement while minimising total particle loss. Particles were carefully transferred to the centre of the ATR crystal using a clean fine-pointed tool. The crystal clamp was tightened to ensure adequate contact and the spectrum acquired with peak labels applied. Spectra were saved in CSV and PDF format. Each spectrum was uploaded to the OpenSpecy platform for comparison against the polymer spectral library, and the best match identity, material class, match value, and signal-to-noise ratio were recorded. An earlier FTIR session on 11 March 2026 was unsuccessful due to particle loss during transfer from filter paper to crystal; the protocol was modified as described above for the final successful session on 21 April 2026.

6. References

- Araujo, C.F., Nolasco, M.M., Ribeiro, A.M.P. and Ribeiro-Claro, P.J.A. (2018) 'Identification of microplastics using Raman spectroscopy: Latest developments and future prospects', *Water Research*, 142, pp. 426-440.
- Barboza, L.G.A., Dick Vethaak, A., Lavorante, B.R.B.O., Lundebye, A.K. and Guilhermino, L. (2018) 'Marine microplastic debris: An emerging issue for food security, food safety and human health', *Marine Pollution Bulletin*, 133, pp. 336-348.
- Cole, M., Lindeque, P., Halsband, C. and Galloway, T.S. (2011) 'Microplastics as contaminants in the marine environment: A review', *Marine Pollution Bulletin*, 62(12), pp. 2588-2597.
- Cowger, W., Steinmetz, Z., Gray, A., Munno, K., Lynch, J., Hapich, H., Primpke, S., De Frond, H., Rochman, C. and Herodotou, O. (2021) 'Microplastic spectral classification needs an open source community: Open Specy to the rescue!', *Analytical Chemistry*, 93(21), pp. 7543-7548.
- Cox, K.D., Covernton, G.A., Davies, H.L., Dower, J.F., Juanes, F. and Dudas, S.E. (2019) 'Human consumption of microplastics', *Environmental Science and Technology*, 53(12), pp. 7068-7074.
- Dawes, C. (2008) 'Salivary flow patterns and the health of hard and soft oral tissues', *Journal of the American Dental Association*, 139(Suppl 2), pp. 18S-24S.
- de Falco, F., Di Pace, E., Cocca, M. and Avella, M. (2019) 'The contribution of washing processes of synthetic clothes to microplastic pollution', *Scientific Reports*, 9(1), p. 6633.
- Dris, R., Gasperi, J., Mirande, C., Mandin, C., Guerrouache, M., Langlois, V. and Tassin, B. (2017) 'A first overview of textile fibres, including microplastics, in indoor and outdoor environments', *Environmental Pollution*, 221, pp. 453-458.
- European Food Safety Authority (EFSA) (2016) 'Presence of microplastics and nanoplastics in food, with particular focus on seafood', *EFSA Journal*, 14(6), p. e04501.
- Fortune Business Insights (2022) Chewing Gum Market Size, Share and COVID-19 Impact Analysis. Available at: <https://www.fortunebusinessinsights.com/chewing-gum-market-102943> (Accessed: 15 March 2026).
- Lowe, M.P. and Koelmans, A.A. (2019) 'Quantification of microplastics released from chewing gum brands under simulated oral conditions', *Environmental Science and Technology Letters*, 6(10), pp. 593-598.
- Jenner, L.C., Rotchell, J.M., Bennett, R.T., Cowen, M., Tentzeris, V. and Sadofsky, L.R. (2022) 'Detection of microplastics in human lung tissue using μ FTIR spectroscopy', *Science of the Total Environment*, 831, p. 154907.
- Käppler, A., Fischer, D., Oberbeckmann, S., Schernewski, G., Labrenz, M., Eichhorn, K.J. and Voit, B. (2016) 'Analysis of environmental microplastics by vibrational microspectroscopy: FTIR, Raman or both?', *Analytical and Bioanalytical Chemistry*, 408(29), pp. 8377-8391.
- Lebling, N. (2018) 'Chewing gum and microplastics: An overlooked source of human exposure', *Environmental Health Perspectives*, 126(9), p. 094001.

- Leslie, H.A., van Velzen, M.J.M., Brandsma, S.H., Vethaak, A.D., Garcia-Vallejo, J.J. and Lamoree, M.H. (2022) 'Discovery and quantification of plastic particle pollution in human blood', *Environment International*, 163, p. 107199.
- Lusher, A., Hollman, P. and Mendoza-Hill, J. (2017) *Microplastics in Fisheries and Aquaculture*. FAO Fisheries and Aquaculture Technical Paper 615. Rome: FAO.
- Morales-Caselles, C., González-Fernández, D., Hanke, G., Arroyo, G.M. and Pérez-Lloréns, J.L. (2021) 'Polymer composition of chewing gum base and microplastic release during mastication', *Marine Pollution Bulletin*, 164, p. 111996.
- Panko, J.M., Chu, J., Harrington, M.E. and Perkins, C. (2013) 'Measurement of airborne concentrations of tire and road wear particles in urban and rural areas of France, Japan, and the United States', *Atmospheric Environment*, 72, pp. 192-199.
- Rochman, C.M. *et al.* (2019) 'Rethinking microplastics as a diverse contaminant suite', *Environmental Toxicology and Chemistry*, 38(4), pp. 703-711.
- Ruiz-Aceituno, L., García-Sarriá, M.J., Lapeña, D., Ramos, L. and Sanz, M.L. (2021) 'Microplastics in chewing gum: Study of the occurrence and characterization in nine commercial chewing gum brands', *Food Chemistry*, 352, p. 129324.
- Thompson, R.C., Swan, S.H., Moore, C.J. and vom Saal, F.S. (2004) 'Our plastic age', *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1526), pp. 1973-1976.
- World Health Organization (WHO) (2022) *Dietary and inhalation exposure to nano- and microplastic particles and potential implications for human health*. Geneva: WHO. Available at: <https://www.who.int/publications/i/item/9789240054608> (Accessed: 20 March 2026).
- Zhang, J., Wang, L. and Kannan, K. (2021) 'Microplastics in house dust from 12 countries and associated human exposure', *Environment International*, 134, p. 105314.

7. Appendices

Appendix A: ATR-FTIR Instrument Peak Data

Tables A1–A5 present the complete peak assignments recorded by the Agilent Technologies Cary 630 ATR-FTIR instrument for each sample, as displayed on the original instrument output reports (Figures 10,12,14,16 and 18). Instrument parameters: 32 sample scans; 32 background scans; resolution 8 cm^{-1} ; range 4000-650 cm^{-1} ; Happ-Genzel apodization; System Status: Good.

Table A1. Sample 1 - Wrigley's Juicy Fruit. Instrument date/time: 21/04/2026 15:22.

Peak No.	Wavenumber (cm^{-1})	Intensity (% T)
1	711.92	77.42
2	872.20	63.19
3	1017.56	69.90
4	1107.02	77.26
5	1230.02	69.12
6	1420.12	60.20
7	1539.39	86.14
8	1576.66	88.64
9	1736.94	80.23
10	2851.41	70.50
11	2918.51	59.57

Table A2. Sample 2 - Wrigley's Apple Extra. Instrument date/time: 21/04/2026 14:49.

Peak No.	Wavenumber (cm^{-1})	Intensity (% T)
1	872.20	93.72
2	1017.56	92.97
3	1103.29	95.05
4	1170.38	95.55
5	1233.75	94.81
6	1420.12	93.53
7	1543.12	97.18
8	1576.66	97.34
9	1736.94	96.32
10	2851.41	95.73
11	2918.51	94.73

Table A3. Sample 3 - Wrigley's Airwaves Extreme. Instrument date/time: 21/04/2026 14:58.

Peak No.	Wavenumber (cm ⁻¹)	Intensity (% T)
1	667.19	86.18
2	875.92	90.11
3	1017.56	84.95
4	1103.29	90.68
5	1177.84	92.09
6	1233.75	90.20
7	1367.93	89.90
8	1423.84	90.69
9	1733.21	92.34
10	2855.14	92.43
11	2922.23	90.67

Table A4. Sample 4 - Wrigley's Extra Refreshers Peppermint. Instrument date/time: 21/04/2026 15:12.

Peak No.	Wavenumber (cm ⁻¹)	Intensity (% T)
1	715.65	81.80
2	872.20	74.09
3	950.47	86.18
4	1021.29	74.67
5	1103.29	80.63
6	1233.75	74.93
7	1386.57	69.57
8	1461.12	71.19
9	1736.94	80.69
10	2851.41	75.12
11	2918.51	63.50
12	2955.78	74.75

Table A5. Sample 5 - Off-brand Watermelon Gum (Primark). Instrument date/time: 21/04/2026 15:17.

Peak No.	Wavenumber (cm ⁻¹)	Intensity (% T)
1	670.92	83.76
2	715.65	79.01
3	872.20	73.30
4	1021.29	73.53
5	1110.75	79.52
6	1233.75	74.48
7	1386.57	68.90
8	1420.12	70.30
9	1736.94	80.19
10	2851.41	76.88
11	2918.51	66.33
12	2955.78	77.17